

Getting Started with React



<https://react.dev/learn>

Objectives

- ◆ Why React mixes markup with rendering logic
- ◆ How JSX is different from HTML
- ◆ How to display information with JSX

Node.js

- ◆ Node.js không phải là một *framework* hay *library*
- ◆ Node.js là môi trường thực thi mã *runtime environment* JavaScript trên máy chủ, không cần trình duyệt
- ◆ Khác với trình duyệt web, nơi JavaScript thường được sử dụng để tương tác với DOM
- ◆ Node.js đi kèm với một số module như *npm* (Node Package Manager) để quản lý các thư viện JavaScript.
- ◆ Node.js được triển khai thông qua framework như *Express.js* để xây dựng ứng dụng web.

Express

- ◆ Express là một **framework** phát triển ứng dụng web trong môi trường **Node.js**
- ◆ Express được thiết kế để hoạt động cùng với Node.js, sử dụng các API của Node.js để quản lý máy chủ.
- ◆ Để sử dụng Express, cần cài đặt Node.js tích hợp Express vào ứng dụng.

React

- ◆ React là một thư viện *JavaScript library* để xây dựng giao diện người dùng – UI cho ứng dụng Web
 - *Component-Based*: tái sử dụng
 - *Virtual DOM*: tạo ra một bản sao ảo của DOM và chỉ cập nhật các phần tử cần thiết
 - *JSX*: cú pháp để viết mã HTML trong mã JavaScript

Next.js

- ◆ Next.js là một *framework React*
- ◆ Next.js được xây dựng dựa trên Node.js
- ◆ Để sử dụng Next.js, cần cài đặt Node.js Node.js sẽ cung cấp môi trường thực thi để chạy mã JavaScript của Next.js

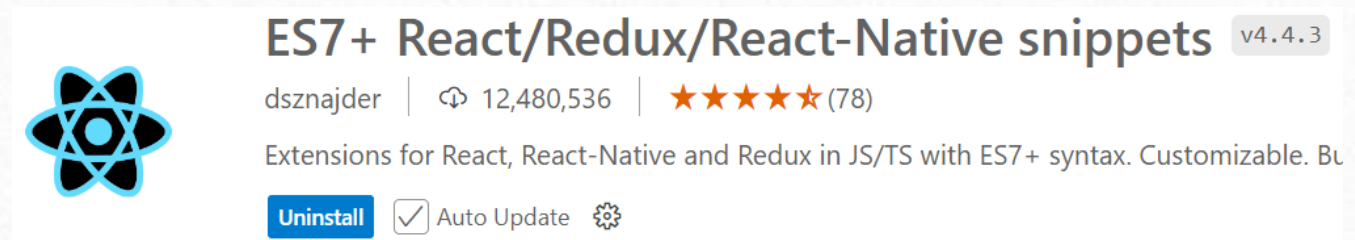
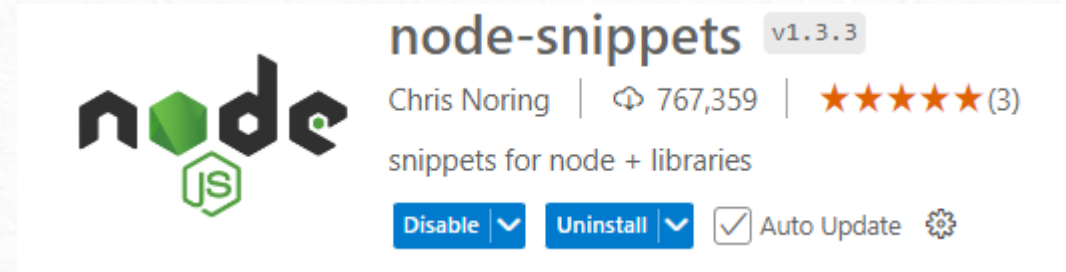
TypeScript



- ◆ <https://www.typescriptlang.org/>
- ◆ TypeScript is **JavaScript with syntax for types**.
- ◆ TypeScript is a strongly typed programming language that builds on JavaScript, giving you better tooling at any scale.

Installation

- ◆ Install *Node.js* for local development
- ◆ Install *Visual Studio Code*
- ◆ Extensions:
 - Node-snippets
 - React-snippets



Snippets

imp→	import moduleName from 'module'
imn→	import 'module'
imd→	import { destructuredModule } from 'module'
imr→	import React from 'react'
imrd→	import ReactDOM from 'react-dom'
imrc→	import React, { Component } from 'react'

rfc

```
import React from 'react'

export default function $1() {
  return <div>$0</div>
}
```

rcc

```
import React, { Component } from 'react'

export default class FileName extends Component {
  render() {
    return <div>$2</div>
  }
}
```

What is JSX?

- ◆ JSX stands for JavaScript XML
- ◆ JSX is a syntax extension for JavaScript.
- ◆ JSX lets you write HTML-like markup inside a JavaScript file.

```
Sidebar() {  
  if (isLoggedIn()) {  
    <p>Welcome</p>  
  } else {  
    <Form />  
  }  
}
```

Sidebar.jsx

```
Form() {  
  onClick() {...}  
  onSubmit() {...}  
  
  <form onSubmit>  
    <input onClick />  
    <input onClick />  
  </form>  
}
```

Form.js

Why JSX?

- ◆ The Web has been built on HTML, CSS, and JavaScript.
 - content in HTML,
 - design in CSS,
 - logic in JavaScript
 - often in separate files
- ◆ But logic increasingly determined content, rendering logic and markup live together in the same place – this is React component

Converting HTML to JSX

```
<h1>Hedy Lamarr's Todos</h1>

<ul>
  <li>Invent new traffic lights
  <li>Rehearse a movie scene
  <li>Improve the spectrum technology
</ul>
```



```
export default function TodoList() {
  return (
    // ???
  )
}
```

JSX Syntax

- ◆ Return a single root element

- `<div>...</div>`
- `<>...</>`

```
return (  
  <div>  
      
  </div>  
);
```

- ◆ Close all the tags

- `<img />` ~~`<img>`~~
- `<li>oranges</li>` ~~`<li>oranges`~~

```
<article>  
  <h1>My First Component</h1>  
  <ol>  
    <li>Components: UI Building Blocks</li>  
    <li>Defining a Component</li>  
    <li>Using a Component</li>  
  </ol>  
</article>
```

- ◆ camelCase most, NOT ALL

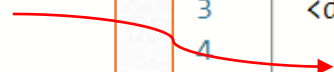
- `strokeWidth`
- `className`

Use a JSX Converter

- ◆ <https://transform.tools/html-to-jsx>

```
1 <!-- Hello world -->
2 <div class="awesome" style="border: 1px solid red">
3   <label for="name">Enter your name: </label>
4   <input type="text" id="name" />
5 </div>
6 <p>Enter your HTML here</p>
```

```
1 <>
2   { /* Hello world */ }
3   <div className="awesome" style={{ border: "1px solid red" }}>
4     <label htmlFor="name">Enter your name: </label>
5     <input type="text" id="name" />
6   </div>
7   <p>Enter your HTML here</p>
8 </>
```



Run JSX

- ◆ Browsers *do not run* JSX directly
- ◆ JSX *not supported* by browsers
- ◆ JSX needs to be *converted* into *standard JavaScript* using *Babel*

What is Babel?



- ◆ **Babel is a JavaScript compiler.** *Use next generation JavaScript, today.*
- ◆ Babel used to convert ECMAScript 2015+ code into a backwards compatible version of JavaScript
- ◆ These plugins allow you to use new syntax, without waiting for browser support
- ◆ Babel can convert JSX syntax
 1. `npm install --save-dev @babel/preset-react`
 2. create `.babelrc` configuration file
 3. `npx babel helloReact.jsx --out-file helloReact.js`

What is React?

- ◆ React is a **JavaScript library** for building user interfaces.
- ◆ React allows us to create **reusable UI components**.
- ◆ React is used to build **single-page applications**.

How does React Work?

- ◆ React creates a **virtual DOM** in memory.
- ◆ When the state of an object **changes**, React **updates** the virtual DOM, then it efficiently updates the **browser DOM** to match the virtual DOM.

How does React Work?

```
src > index.js > ...  
1 import { createRoot } from 'react-dom/client';  
2  
3 function NavBar() {  
4   // TODO: Actually implement a navigation bar  
5   return <h1>Hello from React!</h1>;  
6 }  
7  
8 const domNode = document.getElementById('navigation');  
9 const root = createRoot(domNode);  
10 root.render(<NavBar />);
```

```
public > index.html > ...  
1 <!DOCTYPE html>  
2 <html lang="en">  
3 <head>  
4   <meta charset="UTF-8">  
5   <meta name="viewport" content="width=device-width,&br/>6   <title>Document</title>  
7 </head>  
8 <body>  
9   <div id="root"></div>  
10 </body>  
11 </html>
```

```
<> index.html > ...  
1 <!DOCTYPE html>  
2 <html>  
3   <head><title>My app</title></head>  
4   <body>  
5     <p>This paragraph is a part of HTML.</p>  
6     <nav id="navigation"></nav>  
7     <p>This paragraph is also a part of HTML.</p>  
8   </body>  
9 </html>
```

React and JSX

- ◆ React is a JavaScript *library*
- ◆ JSX is a *syntax* extension
 - JSX and React are two separate things.
 - JSX and React often used together

Library vs Framework

- ◆ library - a collection of functions
- ◆ frameworks - a particular *implementation* of a software application, where your code fills in the details.
- ◆ Hollywood Principle – “**Don’t call us, we’ll call you!**”

Create React App

Create React App makes all the setup for you, including Babel and Webpack configurations.

Create React App

- ◆ Create React App makes all the setup for you, including Babel and Webpack configurations.
- ◆ Creating a new project with npm ≤ 5.2
 - `npm install -g create-react-app`
 - `create-react-app my-app`

Create React App with *npx*

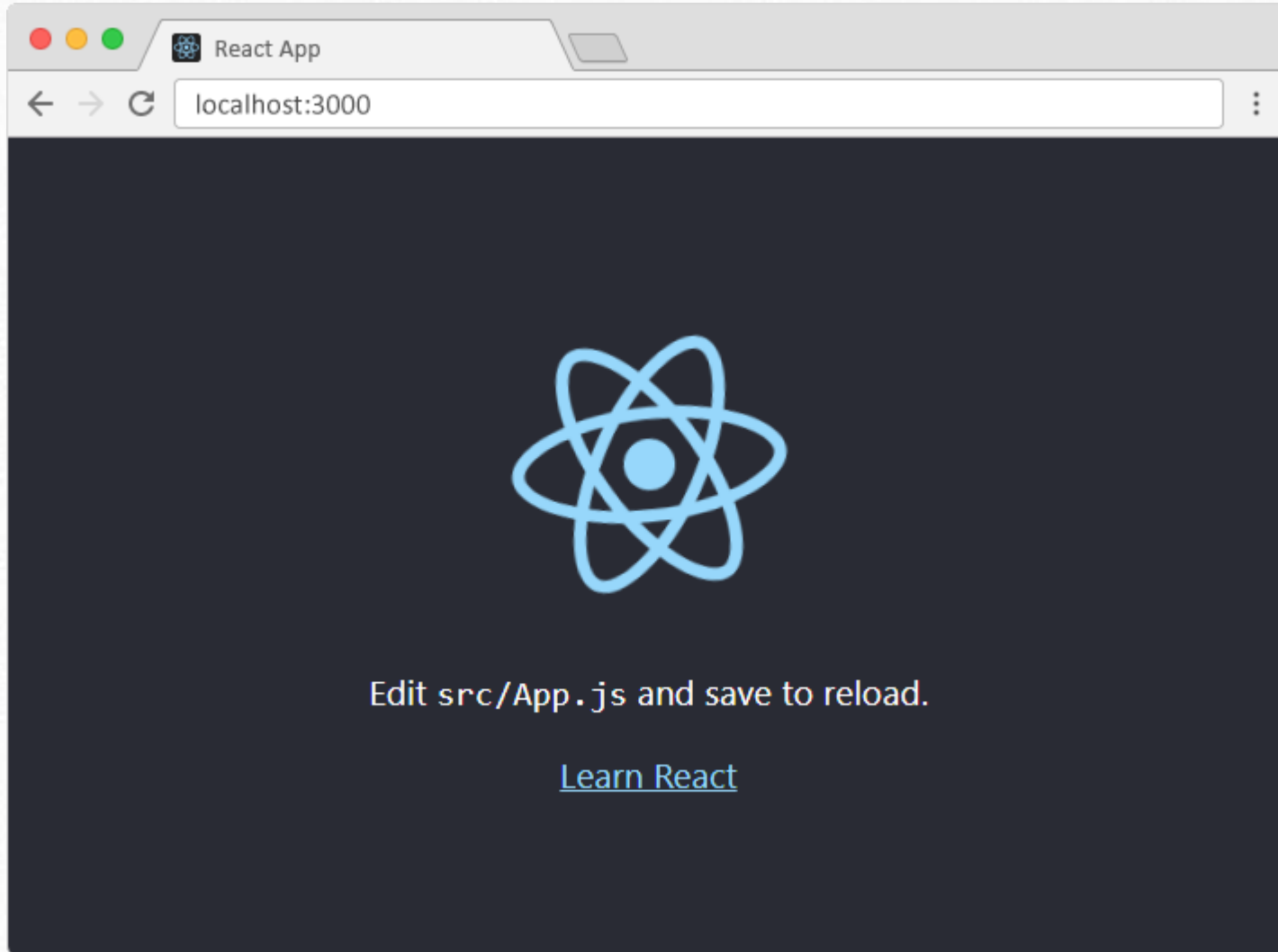
- ◆ *npx* - Node Package eXecute
 - ◆ *npx* automatically installs the latest version of Create React App
 - `npm uninstall -g create-react-app`
 - `npx create-react-app my-app`
- uninstall the package to ensure that **npx** always uses the latest version.

Quick Start

```
npx create-react-app my-app  
cd my-app  
npm start
```

Open <http://localhost:3000/> to see your app

Run the React Application



Folder Structure

```
my-app
├── README.md
├── node_modules
├── package.json
├── .gitignore
├── public
│   ├── favicon.ico
│   ├── index.html
│   ├── logo192.png
│   ├── logo512.png
│   ├── manifest.json
│   └── robots.txt
└── src
    ├── App.css
    ├── App.js
    ├── App.test.js
    ├── index.css
    ├── index.js
    ├── logo.svg
    ├── serviceWorker.js
    └── setupTests.js
```

- ◆ The files must exist with exact filenames:
 - public/index.html
 - src/index.js
- ◆ You can *delete* or *rename* the other files.
- ◆ You may create subdirectories inside *src*

▼ NODEBOX     

▼ public

index.html 

▼ src

 App.js

 index.js

styles.css

 package.json

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1">
6   <title>Document</title>
7 </head>
8 <body>
9   <div id="root"></div>
10 </body>
11 </html>
```

▼ NODEBOX     

▼ public

index.html

▼ src

 App.js

index.js 

styles.css

 package.json

```
1 import React, { StrictMode } from "react";
2 import { createRoot } from "react-dom/client";
3 import "../styles.css";
4
5 import App from "../App";
6
7 const root = createRoot(document.getElementById("root"));
8 root.render(
9   <StrictMode>
10     <App />
11   </StrictMode>
12 );
```

▼ NODEBOX     

▼ public

index.html

▼ src

 App.js 

index.js

styles.css

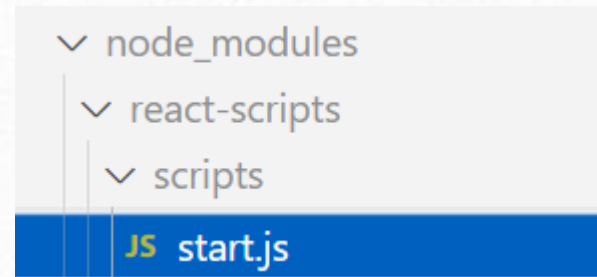
 package.json

```
88
89 export default function App() {
90   return <FilterableProductTable products={PRODUCTS} />;
91 }
92
```

Modify the React Application

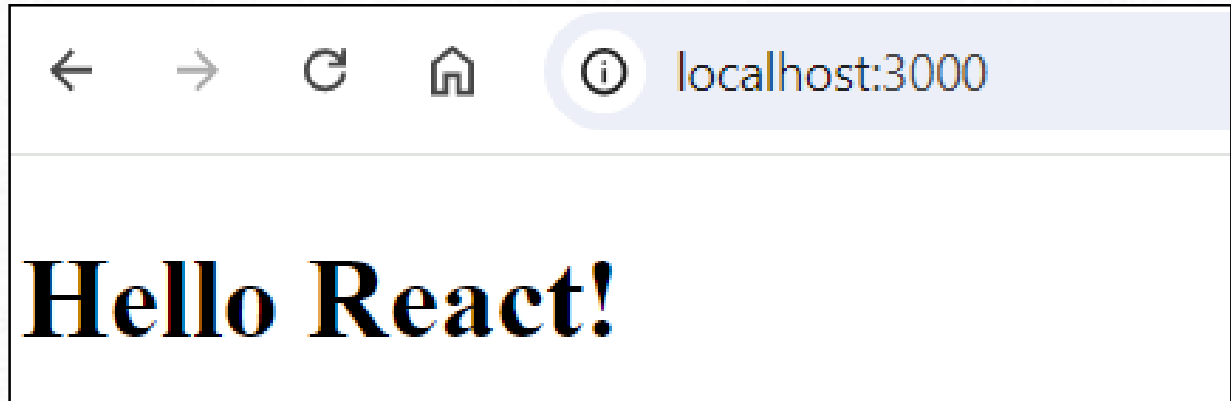
- ◆ npm start → [package.json](#)

```
"scripts": {  
  "start": "react-scripts start",  
}
```



- ◆ [public/index.html](#)
 - root file to load the React application into the browser
- ◆ [src/index.js](#) or [src/App.js](#)
 - initialize the React application and render the application into the DOM element defined in index.html.

Modify index.js



```
JS index.js > ...
import React from 'react';
import ReactDOM from 'react-dom/client';

const myFirstElement = <h1>Hello React!</h1>

const root = ReactDOM.createRoot(document.getElementById('root'));
root.render(myFirstElement);
```

Modify App.js



Hello World!

```
JS index.js > ...
import React from 'react';
import ReactDOM from 'react-dom/client';
import './index.css';
import App from './App';
import reportWebVitals from './reportWebVitals';

const root = ReactDOM.createRoot(document.getElementById('root'));
root.render(
  <React.StrictMode>
    <App />
  </React.StrictMode>
);
reportWebVitals
```

```
function App() {
  return (
    <div className="App">
      <header className="App-header">
        <img src={logo} className="App-logo" alt="logo" />
        <h1>Hello World!</h1>
      </header>
    </div>
  );
}
```

Modify index.html

```
29 <body>
30   <p>This paragraph is a part of HTML.</p>
31   <nav id="navigation"></nav>
32   <p>This paragraph is also a part of HTML.</p>
33   <div id="root"></div>
34 </body>
```



App

```
JS index.js M X
src > JS index.js > ...
1  import React from 'react';
2  import ReactDOM from 'react-dom/client';
3  import './index.css';
4  import App from './App';
5
6  const root = ReactDOM.createRoot(document.getElementById('root'));
7  root.render(<App/>);
8
9  const navigation = ReactDOM.createRoot(document.getElementById('navigation'));
10 navigation.render(<App/>);
```

Exercise

- ◆ Install the create-react-app CLI tool
- ◆ Create a basic React project and served up the compiled project to your browser.